Sequence-aware m6A site prediction and cautious exploration across SG-NEx cell lines

Introduction

N6-methyladenosine (m6A) is an RNA modification that can be investigated using nanopore direct-RNA sequencing. Electrical-event measurements depend on local sequence as well as modification state, complicating prediction. We developed a site-supervised model and applied one frozen pipeline to SG-NEx data [1,2]. Our aim was reproducible candidate prioritisation with transparent limitations, rather than interpreting model scores as measured modification fractions.

Problem statement

Each candidate site contains a seven-base context and a variable-sized bag of reads, each represented by nine measurements: dwell time, event standard deviation (SD) and mean current at three neighbouring events. Labels describe sites, not individual reads. The tasks were to assess site-label prediction and compare predicted patterns across cell lines without confusing missing coverage, technical differences or training exposure with biological differences.

Methods

Data and exploratory rationale. Training data0 comprised 121,838 sites, 5,475 positives and 3,852 genes. Data1 contained 90,810 sites and 6,593 positives; data2 contained 1,323 rows with fractional labels and repeated sequence templates. Only data0 was used for the selected pipeline's fitting and tuning. Data1 had already informed earlier development; data2 was not converted into binary labels.

Exploration showed that pooled central current increased from 114 to 118 between data0 label groups, although positives had lower current within 11/15 adequately supported motifs. This motivated sequence conditioning and distribution summaries. Data0/data1 shared 67,320 literal site keys, including 3,872 label disagreements, supporting cautious transfer interpretation. Valid unusual sites were retained.

Evaluation and model. Early screening included logistic regression, support vector machines, random forest, XGBoost and LightGBM, using 18 read-mean/SD features and shared gene folds. Tuning budgets and verification depth differed; their historical data1-exposed results are not merged into the later comparison.

The later assessment retained the exact five data0 gene folds. For outer fold *f*, inner fold *(f+1) mod 5* selected training duration; fresh weights and training-only normalisation then fitted all four non-assessment folds before scoring *f*. Every gene's sites remained together. Ranking used area under the receiver operating characteristic (ROC) and precision-recall (PR) curves (AUC). Primary score S was the unweighted mean of five (ROC AUC + trapezoidal PR AUC)/2 values. Average precision and pooled out-of-fold areas were separate summaries. Repeated development on these folds limits their interpretation as generalisation estimates.

The selected neural pipeline combined seven-base embeddings, width-48 read encoding, gated attention/mean/SD pooling and 63 all-read distribution summaries. Training sampled at most 128 reads per site, with positive loss weight three; prediction used all valid reads. No probability calibrator was fitted. A fresh 18-epoch model was fitted on all data0, using the median inner-selected duration, and frozen before the reported data1 diagnostic and SG-NEx application.

Results (1): model evaluation

All benchmark rows covered the same 121,838 assessment sites. Logistic regression was a fresh fixed-recipe baseline; LightGBM and the neural model used gene-held-out development fits. Published m6Anet used pretrained HCT116_RNA002 weights: fold IDs stratified its predictions, rather than defining retraining partitions [1].

Pipeline	Mean ROC AUC	Mean PR AUC	Mean S
Logistic regression	0.804056	0.149129	0.476592
LightGBM	0.921897	0.498996	0.710446
Selected cap128 neural model	0.931828	0.505601	0.718715
Published m6Anet reference	0.932176	0.503257	0.717717

The neural-m6Anet S difference was approximately 0.000998; m6Anet had higher ROC AUC. Unresolved pretraining overlap prevents an independent-superiority claim. Increasing the neural training read cap from 32 to 128 increased S from 0.717642 to 0.718715; an added relative-context branch decreased it to 0.708885. Removing positive weighting gave 0.718381 and reduced recall at 0.5. These small development differences do not establish statistical significance.

A separate external neural trial genuinely achieved higher S, 0.726644, improving all five folds. Promotion remained inconclusive: fixed-cutoff/subgroup outcomes were mixed, evidence used one seed and different computational conditions, and its final general prediction interface/artifact was incomplete. At 200+ reads, recall was 151/461 versus 168/461 for cap128 despite better ranking. Retaining the integrated pipeline was a conservative tradeoff, not proof it was globally best.

Frozen data1 ROC/PR/S were 0.827381/0.382350/0.604866, versus LightGBM 0.821764/0.382879/0.602322. Neural recall/F1 were higher at 0.5, whereas LightGBM had higher precision/specificity. This exposed transfer diagnostic did not reselect the model. Final all-data0 predictions were fitted-data outputs, not validation; no binary data2 AUC was claimed.

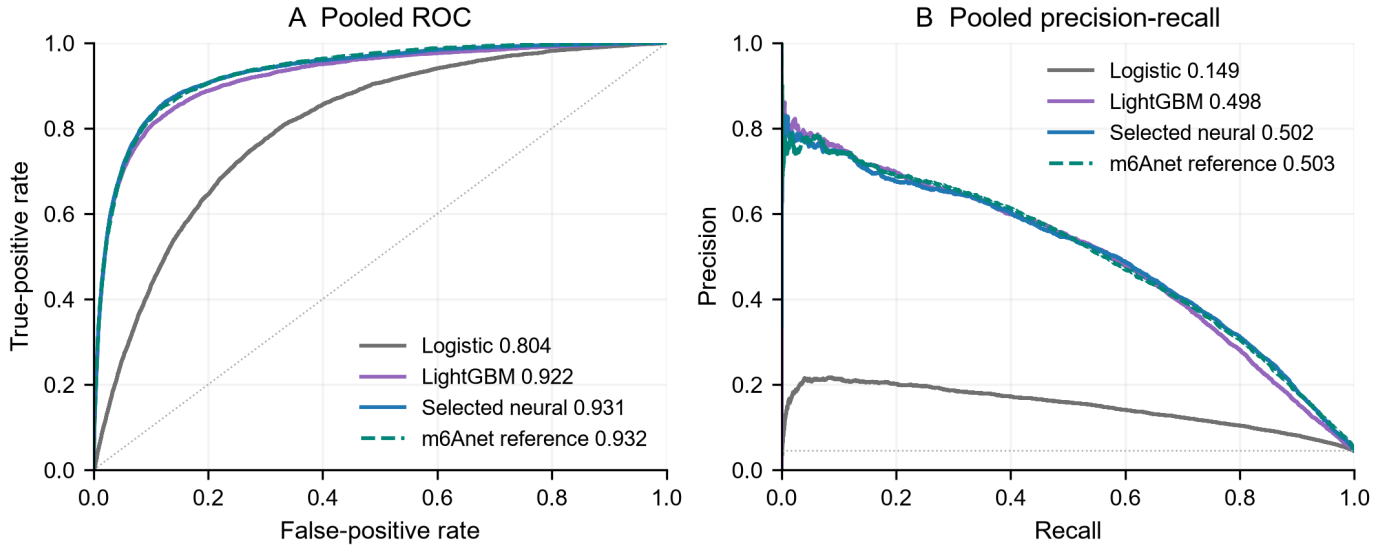


Figure 1. Pooled score curves are supplementary to the mean-fold table; their areas need not coincide. The PR reference reflects 4.49% positive labels. Published m6Anet is a pretrained reference with possible HCT116 overlap, not independent gene-held-out retraining. No significance or calibration claim follows.

Results (2): application to SG-NEx data

SG-NEx comparison. We analysed 23 processed runs spanning seven cell lines and 17 metadata groups. Sites required at least 20 reads for prediction. Matching retained literal transcript positions, explicit per-run aliases and the model's uppercase/U-to-T encoding. Primary comparisons required consistent contexts, annotated human keys and at least two eligible metadata groups per line. Eligible runs were averaged within group, then groups received equal weight. Pairwise and all-seven common populations were separate. Read thresholds 20/50/100 and score cutoffs 0.5/0.7/0.9 were descriptive sensitivities.

The unchanged model produced 2,983,948 scored run-site observations using 232,253,678 reads. Context auditing compared 8,780,071 source observations, including low-read records, at 451,352 scoreable keys: zero context conflicts or exclusions occurred. Another 19,890,009 observations outside that union were counted but not cross-run context-compared. Agreement does not verify historical reference identity or event offsets.

Requiring two groups per line yielded 33,452 all-seven common sites at 20 reads, 11,560 at 50 and 5,058 at 100. On the same 33,452-site panel, mean scores ranged from 0.0705 to 0.0946; 3.76-5.20% met 0.5 and only 15-31 sites per line met 0.9. Hct116 had the highest mean but was also the training cell line. These fractions describe predictions, not modification prevalence.

Median run-pair Spearman agreement at 20 reads was 0.7904 across six same-group pairs, 0.7934 across 28 other-group same-line pairs and 0.7815 across 219 cross-line pairs. Overlapping ranges and shared sequence structure limit a cell-line-separation interpretation. Each correlation uses its own matched sites. Across all 21 line pairs, the exploratory screen required absolute mean-score difference ≥0.2, separated group ranges and coverage ratio ≤2. For example, 2,331/120,195 Hct116-HepG2 shared sites passed; no significance/false-discovery control was applied. All outcomes and valid extremes were retained.

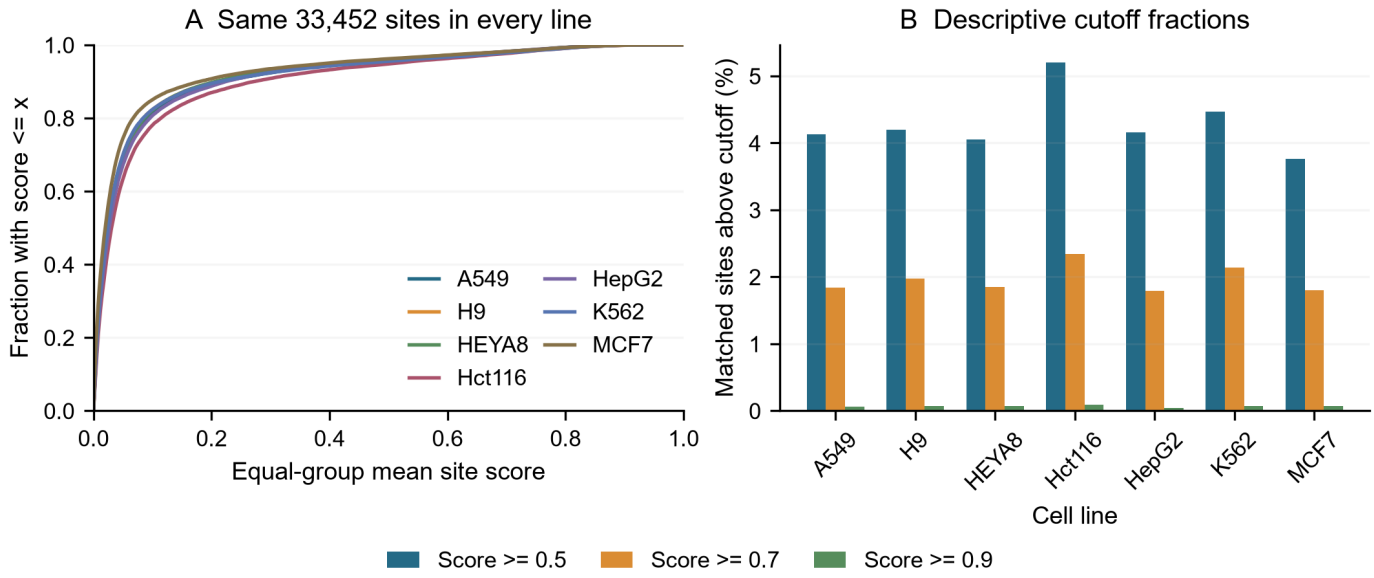
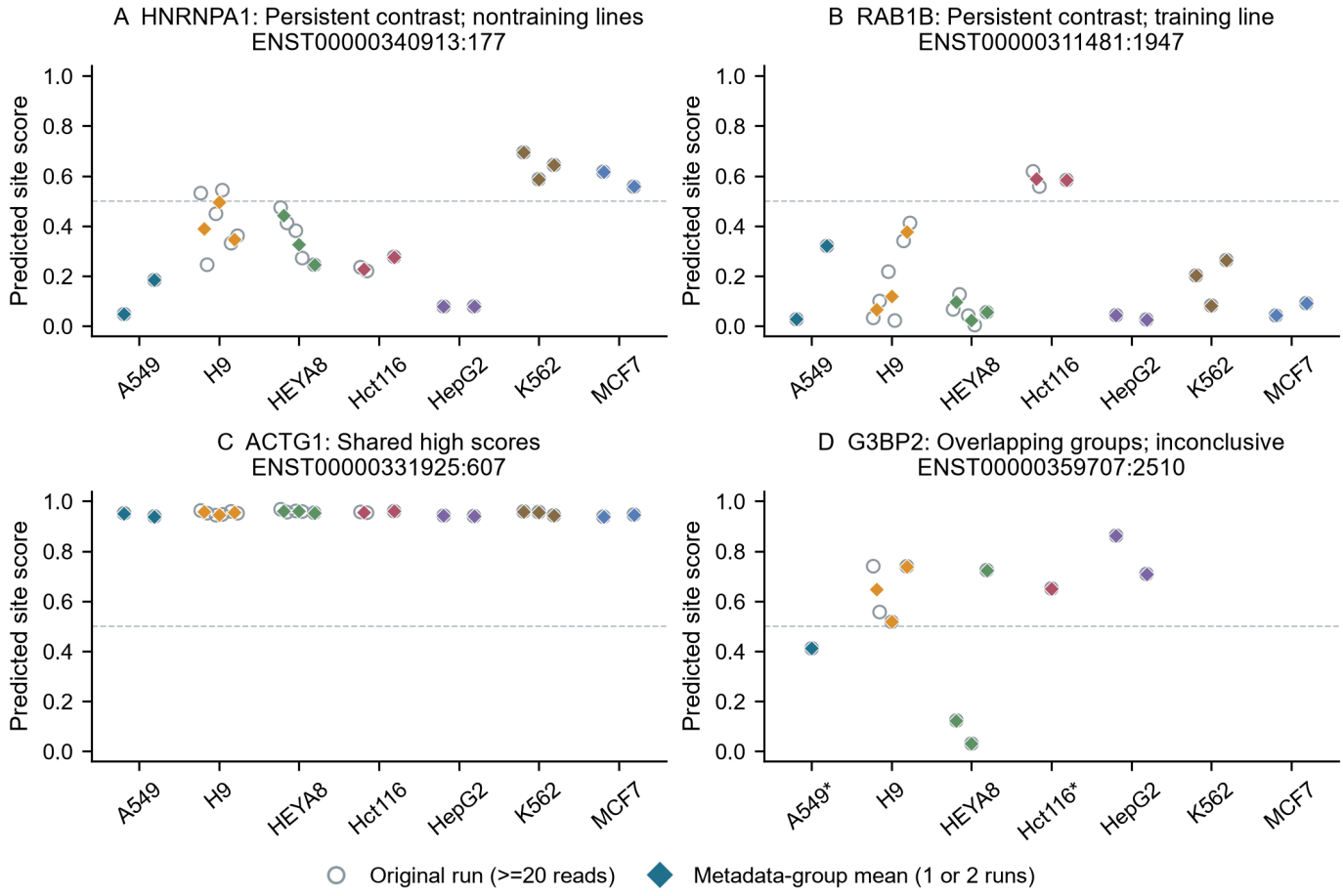


Figure 2. Scores and cutoff fractions use matched, context-qualified all-seven populations with equal group weights. Depth thresholds select different site populations; changes are not a controlled within-site coverage experiment. Cutoffs are descriptive. Hct116's training-domain affinity and technical factors cannot be separated from biology.

Results (2): four illustrative sites

Fixed rules selected four illustrative cases from 1,004 persistent pair-site contrasts representing 545 keys, 41 shared-high sites and 6,098 inconclusive pair-site candidates. HNRNPA1 ENST00000340913:177 showed K562/HepG2 means 0.642/0.078 with separated groups; RAB1B ENST00000311481:1947 showed Hct116/HepG2 means 0.586/0.035. ACTG1 ENST00000331925:607 retained seven-line means 0.942-0.959. G3BP2 ENST00000359707:2510 had overlapping HEYA8/HepG2 group ranges, and its contrast lacked two-group eligibility at 50/100 reads.

Three selected keys occurred in data0, with labels 1/0/1; they are not independent validation. Published HNRNPA1 and ACTG1 assays were not mapped to our selected coordinates [3,4]. Stable threshold results can reuse identical observations. These tail-selected cases prioritise follow-up rather than confirm differential m6A or altered gene function.



Case	7-base context	Positive training labels	Literal key in data0
HNRNPA1	CGGACTG	48	Present; label 1
RAB1B	AGGACAA	41	Present; label 0
ACTG1	TGGACTC	133	Present; label 1
G3BP2	AGGACAT	40	Absent

* One metadata group only (G3BP2 A549/Hct116); blank lines have no >=20-read scores. Training-context support does not establish independent assay validation.

Figure 3. Original run scores and metadata-group summaries retain all four case roles, training exposure and context support. Across 92 run records, 80 were scored and 12 below-20-read records remained unscored. Missing or depth-ineligible is not zero; one-group scores are diagnostic. Group ranges are not confidence intervals, and these selected extremes are not representative.

Discussion

The pipeline provides useful site-label ranking and an inspectable SG-NEx resource, with substantial limits. Among 33,452 common sites, 22,219 had fewer than 20 positive training labels for their seven-base context, including 4,290 with none; all remained with warnings. Reused development folds, exposed data1, uncertain pretrained overlap and one training seed restrict generalisation claims. Basecaller/platform/date differences are confounded with cell line; metadata groups are not verified independent laboratory preparations. Nested depth sensitivities and high score agreement cannot resolve those explanations. Independent compatible assays and verified reference/offset mappings are needed before chemical, stoichiometric or causal interpretation.

Code Availability

Code, the frozen model, example and reproduction instructions are available at LIFO_m6A_project. Hosted explorer: SG-NEx m6A Explorer. The hosted explorer retains all 23 runs and literal case links. Hosted case controls, click/reload navigation and an original-score CSV download passed; anonymous file checks preserved the original data. Native AWS example parity passed; this establishes software portability, not unseen accuracy. Actual new-test input/submission remains separate and unclaimed.

Use of AI methods

Prompts below are summaries; exact chat evidence is retained.

AI tool used	Prompt and output	How output was used
Codex, main project chat	Continue the fixed pipeline; implement/check context QC, comparisons and explorer.	Analysis/UI code, audits, figures and documentation, with saved receipts and independent checks.
Codex, main project chat	Explain findings and prepare an evidence-based report.	Evidence map and this draft for team verification; other members' AI use awaits confirmation.

Two actual project-level corrections were documented. Audit code assumed **fold** and labels existed in the fold CSV; KeyError and inspection revealed **fold_id** without labels. The corrected join passed exact-coverage checks. The assistant also prematurely recommended the higher-S external trial as a clear winner; the user identified its inconclusive promotion review. Rechecking retained its genuine ranking gain while restoring the documented conservative deployment decision.

References

1. Hendra C, Pratanwanich PN, Wan YK, et al. Detection of m6A from direct RNA sequencing using a multiple instance learning framework. *Nature Methods* 19, 1590-1598 (2022). DOI.

2. Chen Y, et al. A systematic benchmark of Nanopore long-read RNA sequencing for transcript-level analysis in human cell lines. *Nature Methods* 22, 801-812 (2025). DOI.

3. Sheehan et al. In situ visualization of m6A sites in cellular mRNAs. *Nucleic Acids Research* 51, e101 (2023). DOI.

4. Mirza et al. SCARPET: site-specific quantification of methylated and nonmethylated adenosines reveals m6A stoichiometry. *RNA* 30, 308-324 (2024). DOI.